

$\zeta(5)$ IS IRRATIONAL

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Abstract

We prove that $\zeta(5)$ is irrational. The proof constructs integer polynomials Q_n of degree $37n$ with $0 < Q_n(\zeta(5)) < \exp(-139n^2/5)$ for all sufficiently large n . These polynomials are rationally normalized Hankel determinants. A positive moment representation gives nonvanishing and, through a logarithmic-energy estimate, bounds their real values. Local estimates for the entire rational functional control the coefficients and show that the normalization preserves the required decay. A perturbation estimate also gives $|\zeta(5) - a/b| > b^{-260}$ for every integer a and all sufficiently large positive integers b .

Keywords. Riemann zeta function, irrationality, Hankel determinant, Bernoulli numbers, logarithmic energy.

1 Introduction

Apéry proved in 1978 that $\zeta(3)$ is irrational [1]. For higher odd arguments, much of the subsequent progress has concerned families of values. Rivoal [10] and Ball and Rivoal [2] proved that infinitely many odd zeta values are irrational. Zudilin [12, 13] showed that at least one of $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ is irrational. Later work has strengthened the lower bounds for the number of irrational values [6] and for the dimension of their rational span [5]. These theorems give information about the family as a whole, but do not determine the irrationality of a specified odd zeta value beyond $\zeta(3)$.

For $\zeta(5)$ itself, Brown and Zudilin [4, Theorem 1] constructed an effective infinite sequence of rational approximations satisfying

$$0 < \left| \zeta(5) - \frac{p}{q} \right| < q^{-0.86}.$$

Their construction uses cellular integrals and the arithmetic of the associated linear forms. The exponent 0.86 does not by itself imply irrationality. To obtain irrationality from approximations of this kind, one needs a decay estimate that remains sufficient after the denominators have been cleared. The same comparison between real decay and denominator growth also governs the polynomial construction used below.

The difficulty of the individual problem is reflected in recent accounts. Lai, Lupu, and Sprang [8, §1] described the irrationality of $\zeta(5)$ as a problem that “seems beyond reach”. Fischler [5, §1] likewise records the absence of an individual result beyond $\zeta(3)$. The present paper proves the next case.

Theorem 1.1. *The number $\zeta(5)$ is irrational.*

The proof uses integer polynomials of growing degree, rather than linear forms in 1 and $\zeta(5)$. They are obtained from Hankel determinants whose entries are affine in the polynomial variable. At $\zeta(5)$, a positive moment representation makes the determinants nonzero. The main estimates show that their values remain small enough to exclude rationality after every coefficient has been made integral. In this way, nonvanishing and the comparison between real decay and denominator growth are established for the same sequence of polynomials.

The construction also gives a bound for rational approximation. This requires a further estimate ensuring that the determinants remain positive at rational arguments sufficiently close to $\zeta(5)$.

Corollary 1.2. *There is a positive integer b_0 such that*

$$\left| \zeta(5) - \frac{a}{b} \right| > b^{-260} \quad (a \in \mathbb{Z}, b \in \mathbb{Z}, b \geq b_0). \quad (1.1)$$

Thus the irrationality exponent of $\zeta(5)$ is at most 260. The threshold b_0 is not specified. The additional estimates and the proof of this corollary are given in Appendix C. They are not needed for Theorem 1.1.

1.1 The construction and the proof

The polynomial estimates needed for irrationality are stated in Theorem 2.1. For all sufficiently large n , they give

$$Q_n := Q_{40n,200} \in \mathbb{Z}[X], \quad \deg Q_n = 37n, \quad 0 < Q_n(\zeta(5)) < \exp\left(-\frac{139}{5}n^2\right).$$

These estimates imply Theorem 1.1 directly. Indeed, suppose that $\zeta(5) = a/b$, where $a \in \mathbb{Z}$ and $b \in \mathbb{Z}_{>0}$. Then

$$0 < b^{37n} Q_n(a/b) < \exp\left(37n \log b - \frac{139}{5}n^2\right).$$

The middle term is an integer and the right-hand side tends to zero. The quadratic decay therefore suffices, without an estimate for the heights of the polynomials. The rest of the proof is devoted to constructing polynomials with these properties.

We begin with an $h \times h$ Hankel matrix $G_K(X)$, where $h = 37K/40$, whose entries are values of a rational functional. Each entry is affine in X . At $X = \zeta(5)$, the functional becomes integration against a positive weight, and $G_K(\zeta(5))$ is a positive moment matrix. Its determinant is consequently positive. A multiple-integral representation expresses that determinant in terms of a squared Vandermonde factor and the weight. After scaling, the logarithm of the integrand is estimated by a logarithmic-energy inequality. This gives the required real bound on the K^2 scale.

The arithmetic estimates must hold for every coefficient of the determinant, not only for its value at $\zeta(5)$. They must also account for the whole rational functional. Polynomial division produces both polynomial quotients and simple-pole terms, and an estimate based only on favorable residues would leave the former uncontrolled. We therefore retain both contributions in the local estimates over $\mathbb{Q}_p[X]$.

The main determinant estimates take two forms. For the intermediate primes, a basis with prescribed vanishing at the pole classes gives a determinant valuation. For the larger primes,

another basis isolates the possible negative valuations of the residue terms. The remaining polynomial contribution is handled by a rank bound. Together with a coarse bound for the smaller primes, these estimates give a positive rational normalization that turns the determinant into the integer polynomial $Q_{K,M}$. The cutoff M is kept fixed when the local estimates are summed. The cost of this normalization is smaller than the real decay. Choosing $M = 200$ and $K = 40n$ gives the polynomials Q_n above.

For Corollary 1.2, smallness at $\zeta(5)$ alone is not enough, since the polynomial could vanish at a nearby rational argument. We bound the coefficient matrix of X relative to the positive matrix $G_K(\zeta(5))$. This preserves positivity under the required perturbation and allows the integer argument to be applied at the rational point. The larger fixed cutoff used in the appendix provides the additional decay for that argument.

1.2 Organization

Section 2 defines the matrix and proves its degree and positivity. Sections 3 and 4 establish the local estimates, which are summed in Section 5. The real bound is proved in Section 6, and the two bounds are combined in Section 7. Appendices A and B contain the rational data and elementary estimates for the constants. Appendix C proves the approximation bound and records a coefficient height estimate.

2 The determinant construction

We first define a rational functional whose value at $X = \zeta(5)$ is integration against a positive weight. Applying it to rational functions with a common denominator gives the Hankel matrix. The residues determine the degree of its determinant, while the integral representation proves positivity.

2.1 The functional and the matrix

Throughout the paper,

$$K = 40n, \quad N = 3n, \quad h = 37n, \quad \alpha = \frac{3}{40}, \quad \lambda = \frac{37}{40}, \quad H = 1 + 2\alpha = \frac{23}{20}. \quad (2.1)$$

Set $D_0 = 1$ and

$$D_m(t) = \prod_{j=1}^m (t + j^2), \quad H_j^{(5)} = \sum_{v=1}^j \frac{1}{v^5}, \quad H_0^{(5)} = 0.$$

We use the convention $B_1 = -1/2$ for the Bernoulli numbers. Define a $\mathbb{Q}$ -linear functional with values in $\mathbb{Q}[X]$ by

$$\mu_X(t^e) = \frac{(-1)^e B_{2e+2} (2e+3)(2e+4)(2e+5)}{24}, \quad e \geq 0, \quad (2.2)$$

$$\mu_X\left(\frac{1}{t+j^2}\right) = j^4(X - H_j^{(5)}) - \frac{1}{4} + \frac{1}{2j}, \quad j \geq 1. \quad (2.3)$$

Polynomial division and simple partial fractions extend μ_X to rational functions with simple poles among the negative integer squares. We write μ for its restriction to polynomials. Only the simple-pole values depend on X , and that dependence is affine.

Define

$$G_K(X) = \left[\mu_X \left(\frac{D_N(t)^6 t^{i+j}}{D_K(t)} \right) \right]_{0 \leq i, j < h}, \quad \Delta_K(X) = \det G_K(X), \quad (2.4)$$

and

$$S_K = \frac{(K!)^{2h} 4^{h-1}}{(N!)^{12h} \prod_{i=1}^{h-1} ((2i)!)^2}, \quad F_K = S_K \Delta_K. \quad (2.5)$$

For a fixed integer $M \geq 40$, Section 5 defines a positive rational number $m_{K,M}$ whenever $K \geq 200M^2$. The polynomial used below is

$$Q_{K,M}(X) = m_{K,M} F_K(X). \quad (2.6)$$

The factor $m_{K,M}$ will be specified by prime exponents, without computing the content of Δ_K .

2.2 The polynomial estimates

The estimates required in the introduction are collected below. The cutoff M is always fixed before n tends to infinity.

Theorem 2.1. *For each integer $M \geq 40$ and each $K \in 40\mathbb{Z}_{>0}$ with $K \geq 200M^2$, the polynomial $Q_{K,M}$ belongs to $\mathbb{Z}[X]$, has degree h , and satisfies $Q_{K,M}(\zeta(5)) > 0$. For all sufficiently large positive integers n ,*

$$0 < Q_{40n,200}(\zeta(5)) < \exp\left(-\frac{139}{5}n^2\right), \quad (2.7)$$

$$0 < Q_{40n,100000}(\zeta(5)) < \exp\left(-\frac{7907}{100}n^2\right). \quad (2.8)$$

The choice $M = 200$ suffices for irrationality. The larger cutoff gives the additional decay used in Appendix C. Neither eventual threshold is specified. We prove degree and positivity here, integrality in Section 5, and decay in Section 7.

2.3 Degree

Let $D_{\text{tail}} = D_K/D_N$ and $W = D_N^5$. After cancellation, the rational functions in (2.4) have exactly the possible poles $-j^2$ with $N < j \leq K$. If $V_{ij} = (-j^2)^i$, then

$$[X]G_K(X) = V \operatorname{diag} \left(\frac{j^4 D_N(-j^2)^5}{D'_{\text{tail}}(-j^2)} \right)_{N < j \leq K} V^\top.$$

The Vandermonde identity therefore gives

$$[X^h]\Delta_K(X) = (-1)^{h(h-1)/2} \prod_{j=N+1}^K j^4 D_N(-j^2)^5 \neq 0. \quad (2.9)$$

In particular, all the polynomials in (2.6) have degree h once their normalizations have been defined.

2.4 The positive integral representation

To prove nonvanishing at $\zeta(5)$, we realize the matrix as a positive Gram matrix. The use of positive integrals in irrationality arguments is familiar from Beukers' proofs for $\zeta(2)$ and $\zeta(3)$ [3]. The representation below will also give the real bound in Section 6.

Proposition 2.2. *For every rational function R in the domain of μ_X ,*

$$\mu_{\zeta(5)}(R) = \int_0^\infty R(y^2)w(y) dy, \quad w(y) = \frac{(2\pi)^4 y^5}{12} \sum_{\ell \geq 1} \ell^4 e^{-2\pi \ell y}. \quad (2.10)$$

Hence $G_K(\zeta(5))$ is positive definite.

Proof. The weight is positive. It tends to $1/\pi$ as $y \downarrow 0$ and is $O((1+y)^5 e^{-2\pi y})$ at infinity. Euler's formula for the even zeta values [9, (25.6.2)] gives

$$\int_0^\infty y^{2e} w(y) dy = \frac{(2e+5)!}{12(2\pi)^{2e+2}} \zeta(2e+2) = \mu(t^e).$$

For a pole, put $f(y) = (e^{2\pi y} - 1)^{-1}$. Then $w(y) = y^5 f^{(4)}(y)/12$, and direct differentiation gives

$$\left(\frac{d}{dy}\right)^4 \frac{y^5}{y^2 + a^2} = 24a^4 \operatorname{Im}(a - iy)^{-5}.$$

Four integrations by parts have vanishing boundary terms. At zero each boundary product is $O(y)$, and at infinity it decays exponentially. Hermite's integral formula [9, (25.11.29)], with $a > 0$, yields

$$\int_0^\infty \frac{w(y)}{y^2 + a^2} dy = a^4 \zeta(5, a) - \frac{1}{2a} - \frac{1}{4}.$$

At $a = j \in \mathbb{Z}_{>0}$ this is $j^4(\zeta(5) - H_j^{(5)}) - 1/4 + 1/(2j)$. Linearity proves (2.10).

If $q \neq 0$ is a real polynomial of degree less than h , then

$$\int_0^\infty \frac{D_N(y^2)^6 q(y^2)^2}{D_K(y^2)} w(y) dy > 0.$$

This is the quadratic form of $G_K(\zeta(5))$ at the coefficient vector of q . □

3 The local rational functional

The arithmetic estimates are made with X indeterminate, so that they apply to every coefficient of the determinant. Under $t = -x^2$, the poles become integers and Bernoulli multiplication can be applied by residue class. We extend the resulting functional to convergent p -adic expansions, prove its distribution formula, and obtain a separate bound for the small primes.

For a polynomial $A \in \mathbb{Q}_p[X]$, let $v_p^G(A)$ be the minimum of the p -adic valuations of its coefficients, with $v_p^G(0) = +\infty$. In particular, a lower bound for $v_p^G(A)$ applies to the constant coefficient as well as to the coefficients of positive degree.

Let $L(x^k) = B_k$ and $\tau(P) = L(P''')/24$. For an integer r , put

$$d(r) = \begin{cases} r, & r \geq 0, \\ -r - 1, & r < 0, \end{cases} \quad \tau_X\left(\frac{1}{x - r}\right) = H_{d(r)}^{(5)} - X.$$

Extend τ_X by polynomial division and simple partial fractions. The pullback identity is

$$\mu_X(R) = \tau_X(x^5 R(-x^2)). \quad (3.1)$$

For monomials this follows from three differentiations. For a pole, use

$$\frac{x^5}{j^2 - x^2} = -x^3 - j^2 x - \frac{j^4}{2} \left(\frac{1}{x - j} + \frac{1}{x + j} \right)$$

and $H_j^{(5)} - H_{j-1}^{(5)} = j^{-5}$. The reflection and difference identities for Bernoulli polynomials [9, (24.4.1), (24.4.3)] give

$$\tau_X(g(-1 - x)) = -\tau_X(g(x)), \quad (3.2)$$

$$\tau_X(g(x + 1) - g(x)) = \frac{g^{(4)}(0)}{24}, \quad (3.3)$$

where (3.3) requires g to be regular at zero. The same identities for simple poles follow from the harmonic increments.

3.1 Completion at a prime

After a residue-class substitution, poles outside $\mathbb{Z}_p$ are represented by convergent power series. We therefore need a functional on both analytic functions and principal parts. Its analytic part can lose one power of p . In the next lemma, the degree restriction prevents this loss in the initial term, and the factors p^j absorb it in the remaining terms.

Fix $p \geq 7$. For $d \geq 3$, write

$$\kappa_d = \frac{d(d-1)(d-2)B_{d-3}}{24}, \quad \kappa_0 = \kappa_1 = \kappa_2 = 0.$$

The von Staudt–Clausen theorem [9, §24.10(i)] implies $v_p(\kappa_d) \geq -1$. Moreover $\kappa_d \in \mathbb{Z}_p$ for $d \leq p + 1$. Thus τ extends continuously to the Tate algebra $\mathcal{A} = \mathbb{Q}_p\langle z \rangle$ by

$$\tau \left(\sum_{d \geq 0} f_d z^d \right) = \sum_{d \geq 0} f_d \kappa_d, \quad \|\tau\| \leq p.$$

The norm on $\mathcal{A}$ is the coefficient supremum norm.

Lemma 3.1. *Let $T(z) = \prod_{\nu=1}^t (z - r_\nu)$, where the r_ν are distinct integers, $r_\nu - r_\eta \in \mathbb{Z}_p^\times$ for $\nu \neq \eta$, and $d(r_\nu) < p$. Let $Y \in \mathbb{Z}_p[X]$. For polynomials $U_j \in \mathbb{Z}_p[z]$ with $\deg U_0 \leq p + 1$,*

$$\tau_Y^{\text{ext}} \left(\sum_{j \geq 0} \frac{p^j U_j(z)}{T(z)} \right) \in \mathbb{Z}_p[X]. \quad (3.4)$$

Proof. Polynomial division and partial fractions give

$$\frac{U}{T} = P + \sum_{\nu=1}^t \frac{c_\nu}{z - r_\nu}, \quad c_\nu = \frac{U(r_\nu)}{T'(r_\nu)}.$$

This map has norm at most one from $\mathbb{Z}_p[z]$ to the direct sum of the polynomial coefficient space and the residue spaces. It therefore extends to

$$\mathcal{B}_T = \mathcal{A} \oplus \bigoplus_{\nu=1}^t \mathbb{Q}_p \frac{1}{z - r_\nu}$$

with the maximum norm. The resulting decomposition is unique, since an analytic function cannot cancel a nonzero principal part.

On $\mathcal{B}_T$, define

$$\tau_Y^{\text{ext}} \left(f + \sum_{\nu} \frac{c_\nu}{z - r_\nu} \right) = \tau(f) + \sum_{\nu} c_\nu (H_{d(r_\nu)}^{(5)} - Y).$$

The principal-part values are integral and the analytic part loses at most one power of p . In the $j = 0$ term the polynomial quotient has degree at most $p + 1$, so it too has integral value. For $j \geq 1$, the factor p^j absorbs the possible loss. The series converges in $\mathcal{B}_T$, which proves (3.4). $\square$

If $v_p(s) < 0$, expansion in $\mathcal{A}$ gives

$$\tau^{\text{an}} \left(\frac{1}{z - s} \right) = -\frac{1}{4} \sum_{k \geq 0} \binom{k+3}{3} B_k s^{-k-4}. \quad (3.5)$$

A rational function with the fixed near poles r_ν and any far poles has the same value whether its far factors are expanded first or global partial fractions are taken first. This follows from the uniqueness of the decomposition in $\mathcal{B}_T$ and (3.5).

3.2 Distribution

We choose the new parameter to make the distribution formula hold at $1/x$. Translation and reflection will then determine it at every integer pole. Set

$$C_p = \sum_{a=1}^{p-1} \tau^{\text{an}} \left(\frac{1}{x + a/p} \right), \quad Y = p^5 X + C_p. \quad (3.6)$$

The constant C_p belongs to $p^5 \mathbb{Z}_p$. In fact, the $k = 0$ term of (3.5) contains $p^4 \sum_{a=1}^{p-1} a^{-4}$, which is divisible by p^5 . The $k = 1$ term has valuation at least 5, and the terms with $k \geq 2$ have valuation at least $k + 3$.

Lemma 3.2. *For every rational function g with polynomial part and simple integer poles,*

$$\tau_X(g) = p^{-4} \sum_{a=0}^{p-1} \tau_Y^{\text{ext}}(g(a + px)). \quad (3.7)$$

Integer poles on the right use τ_Y , and far poles use (3.5).

Proof. For polynomials, (3.7) follows from Bernoulli multiplication [9, (24.4.18)] after three differentiations. At $g(x) = 1/x$, its right-hand side is $p^{-5}(-Y + C_p) = -X$.

Let $\mathcal{T}(g)$ denote the right-hand side. The substitutions $a \mapsto p - 1 - a$ and $x \mapsto -1 - x$ show that $\mathcal{T}$ satisfies (3.2). For g regular at zero, its translation difference telescopes to

$$\mathcal{T}(g(x+1) - g(x)) = p^{-4} \tau_Y^{\text{ext}}(g(p(x+1)) - g(px)) = \frac{g^{(4)}(0)}{24}.$$

The reflection and difference identities extend to analytic far-pole terms by continuity. Starting at $1/x$, the difference identity determines the values at every positive integer pole. Reflection gives the negative integer poles. This proves the lemma. $\square$

3.3 Small primes

For small primes, a bound independent of the pole classes is sufficient. We use the binomial basis for integer-valued polynomials. If $A(\mathbb{Z}_p) \subset \mathbb{Z}_p$ and $\deg A \leq d$, then the coefficients in its expansion in $\binom{x}{k}$ are integral. Indeed, they are the forward differences of A at zero. Vandermonde's identity then gives

$$v_p(A(x) - A(y)) \geq v_p(x - y) - \left\lfloor \log_p \max(1, d) \right\rfloor \quad (x, y \in \mathbb{Z}_p). \quad (3.8)$$

More generally, its r th derivative is integer-valued up to a loss of $r \left\lfloor \log_p \max(1, d) \right\rfloor$. To see this, expand $\binom{x+z}{k}$ by Vandermonde's identity. A coefficient of degree r in $\binom{z}{j}$ is a sum of products of r reciprocals of integers at most j .

The generating identity

$$L((1+z)^x) = \frac{\log(1+z)}{z}$$

gives $L(\binom{x}{k}) = (-1)^k / (k+1)$. Consequently, if $P(\mathbb{Z}_p) \subset p^{-A} \mathbb{Z}_p$ and $\deg P \leq d$, then

$$v_p(\tau(P)) \geq -A - 4 \left\lfloor \log_p(d+1) \right\rfloor - v_p(24). \quad (3.9)$$

Lemma 3.3. *Let A be integer-valued on $\mathbb{Z}_p$, with $\deg A \leq d$, and put*

$$g(x) = \frac{(K!)^2 A(x)}{\prod_{\substack{-K \leq r \leq K \\ r \neq 0}} (x - r)}.$$

Then

$$v_p^G(\tau_X(g)) \geq -6 \left\lfloor \log_p \max(2K, d+1) \right\rfloor - v_p(24). \quad (3.10)$$

Proof. Write $g = P + \sum_{r \neq 0} c_r / (x - r)$ and let $L_0 = \left\lfloor \log_p(2K) \right\rfloor$. The residues satisfy

$$c_r = \pm \frac{(K!)^2 r A(r)}{(K+r)!(K-r)!}, \quad v_p(c_r) \geq -L_0.$$

At each prime-power level, the difference between the two factorial floor sums is at most one.

Outside the balls $r + p^{L_0+1} \mathbb{Z}_p$, counting roots modulo each p^j shows that $v_p(g(x)) \geq -2L_0$. The same bound holds for all the pole terms. Within such a ball, its center is the only root at distance of valuation greater than L_0 . Put

$$F_r(x) = \frac{(K!)^2}{\prod_{s \neq r} (x - s)},$$

where the product is over the nonzero roots. The inverse-product expansion gives

$$v_p(F_r(x)) \geq -L_0, \quad v_p(F_r(x) - F_r(r)) \geq v_p(x - r) - 2L_0.$$

Subtracting the principal part of $A(x)F_r(x)/(x - r)$ and using (3.8) yields

$$P(\mathbb{Z}_p) \subset p^{-L_0-M_0}\mathbb{Z}_p, \quad M_0 = \max(L_0, \lfloor \log_p \max(1, d) \rfloor).$$

Continuity includes the centers of the balls. Apply (3.9) to P . For the pole terms use $v_p(H_{d(r)}^{(5)}) \geq -5 \lfloor \log_p K \rfloor$. Both bounds imply (3.10). $\square$

We apply Lemma 3.3 in a basis that becomes integer-valued under $t = -x^2$. Define

$$q_0(t) = 1, \quad q_i(t) = \frac{(-1)^i 2t D_{i-1}(t)}{(2i)!} \quad (i \geq 1), \quad f_i(t) = \left(\frac{D_N(t)}{(N!)^2} \right)^3 q_i(t).$$

For $i \geq 1$, the identity

$$q_i(-x^2) = \binom{x+i}{2i} + \binom{x+i-1}{2i}$$

shows that $q_i(-x^2)$ is integer-valued. The same is true of $q_0 = 1$. Together with

$$\frac{D_N(-x^2)}{(N!)^2} = \binom{N-x}{N} \binom{N+x}{N},$$

this proves that $f_i(-x^2)$ is integer-valued for every $i \geq 0$. The triangular change of basis gives

$$F_K(X) = \det \left[(K!)^2 \mu_X \left(\frac{f_i(t)f_j(t)}{D_K(t)} \right) \right]_{0 \leq i, j < h}. \quad (3.11)$$

The pullback numerator has degree at most $12N + 4h + 1$. Thus $\max(2K, d + 1) \leq 5K$, and Lemma 3.3 proves

$$v_p^G(F_K) \geq -6h \lfloor \log_p(5K) \rfloor - h v_p(24) \quad \text{for every prime } p. \quad (3.12)$$

4 Local estimates for the determinant

Applying (3.12) at every prime would give too large a normalizing factor. We fix $M \geq 40$ and improve the estimate for $p > K/M$. Two choices of basis are needed. In the inner range $K/M < p \leq K/3$, the basis distributes vanishing among the pole classes. In the outer range $p > K/3$, it separates the residue terms from a polynomial correction of bounded rank.

4.1 The inner range

Fix $M \geq 40$ and assume

$$K \in 40\mathbb{Z}_{>0}, \quad K \geq 200M^2, \quad K/M < p \leq K/3. \quad (4.1)$$

Then $p > 200M$ and $p^2 > 5K$. For $1 \leq a \leq (p-1)/2$, let

$$\ell_A(a) = \# \{1 \leq j \leq A \mid j \equiv a \text{ or } -a \pmod{p}\}, \quad m_A = \lfloor A/p \rfloor.$$

If a row polynomial has a zero of order $\nu_i(a)$ at $t = -a^2$, substitution $x = c + pz$ in (3.1), for $c = a, p-a$, gives

$$\nu_i(a) + \nu_j(a) + 6\ell_N(a) - \ell_K(a) - 4 \quad (4.2)$$

as a lower valuation for that summand of (3.7). At $c = 0$ the corresponding bound is

$$2\nu_i(0) + 2\nu_j(0) + 12m_N - 2m_K + 1. \quad (4.3)$$

We justify these bounds with their degree restrictions below.

The bounds depend on the vanishing at each class. We assign L_a basis elements to the class $t = -a^2$, with total dimension h , and choose the L_a so that no residue class gives a smaller bound than the assigned row weights.

Put $L_0 = 4M + 10$, $m = (p-1)/2$, and $b_a = 3\ell_N(a)$. Define integers T, E by

$$mT + E = h - L_0 + 3(N - m_N), \quad 0 \leq E < m. \quad (4.4)$$

Give $\epsilon_a = 1$ to the first E classes in decreasing order of $\ell_K(a)$ and give it the value zero elsewhere. Ties may be ordered arbitrarily. Set

$$L_a = T - b_a + \epsilon_a, \quad Z_a = L_a + b_a = T + \epsilon_a.$$

The dimensions L_a are nonnegative and $L_0 + \sum_a L_a = h$. To prove nonnegativity, put $x = K/p$. From (4.4),

$$2Hx - \frac{21}{20} < T < 2Hx, \quad b_a \leq 6\alpha x + 3.$$

Here $(2L_0 + 6m_N)/(p-1) < 1/20$ follows from (4.1). Hence

$$T - b_a > 2\lambda x - \frac{81}{20} \geq \frac{3}{2}.$$

For $a = 0, 1, \dots, m$ and $0 \leq i < L_a$, take the row polynomial

$$E_{a,i}(t) = \prod_{\substack{0 \leq c \leq m \\ c \neq a}} (t + c^2)^{L_c} (t + a^2)^i. \quad (4.5)$$

The factors $t + c^2$ are pairwise coprime modulo p . In the Chinese remainder decomposition by their powers, the polynomials in (4.5) give triangular local bases whose diagonal entries are units. They therefore form a $\mathbb{Z}_p$ -unimodular basis of the polynomials of degree less than h .

Assign the row weights

$$w_{a,i} = i + b_a - \frac{\ell_K(a) + 4}{2}, \quad a > 0, \quad (4.6)$$

$$w_{0,i} = \min \left(2i + 6m_N - m_K + \frac{1}{2}, \min_{1 \leq c \leq m} \left(Z_c - \frac{\ell_K(c) + 4}{2} \right) \right). \quad (4.7)$$

Define

$$\gamma_p^{\text{in}} = 2 \sum_{a=0}^m \sum_{i=0}^{L_a-1} w_{a,i}. \quad (4.8)$$

This is an integer.

Proposition 4.1. *Under (4.1), $v_p^G(\Delta_K) \geq \gamma_p^{\text{in}}$.*

Proof. At $x = c + pz$, every near denominator factor supplies a factor p and a simple integer pole in z . The near poles have absolute sizes at most $M + 1$, and distinct near poles differ by a p -adic unit. Their harmonic indices are less than p . Far factors are units times convergent series in pz .

After the indicated power of p is extracted, the ordinary summand has the form in Lemma 3.1. Its leading numerator has degree at most

$$\nu_i(c) + \nu_j(c) + 6\ell_N(c) \leq 2(T + 1) < \frac{23}{5}M + 2.$$

For the zero summand this degree is at most

$$5 + 2\nu_i(0) + 2\nu_j(0) + 12m_N \leq 5 + 4L_0 + 12m_N \leq \frac{169}{10}M + 45.$$

Both degrees are at most $p + 1$. Lemmas 3.1 and 3.2 therefore prove (4.2) and (4.3) for the entire summands.

It remains to check each assigned weight against every source in the distribution formula. At an ordinary source c , a row from another ordinary block a has half-weight $Z_c - (\ell_K(c) + 4)/2$. This is at least the assigned weight because

$$Z_c - Z_a + 1 + \frac{\ell_K(a) - \ell_K(c)}{2} \geq 0.$$

The counts ℓ_K take two consecutive values. In the only case $Z_a = T + 1$, $Z_c = T$ that needs an ordering, the definition of the extras gives $\ell_K(a) \geq \ell_K(c)$. Formula (4.7) treats the zero-block rows at every ordinary source.

At the zero source, the half-weight of any ordinary row is $2L_0 + 6m_N - m_K + 1/2 \geq 7M + \frac{41}{2}$, larger than every ordinary assigned weight. The remaining comparisons follow directly from (4.7). Thus each entry in the new basis has valuation at least the sum of its two row weights. Every determinant term has valuation at least $2 \sum w_{a,i}$. The basis change is unimodular, which proves the proposition. $\square$

4.2 The outer range

For $p > K/3$, the small number of poles in each class permits a more direct estimate. We first remove the nonintegral polynomial moments, recording an explicit bound for the rank of their contribution. Suppose that

$$p \geq 7, \quad p \leq K < 3p, \quad p^2 > 2K, \quad 2N < p, \quad 5N \leq 2p - 2. \quad (4.9)$$

These conditions hold when $K \geq 200M^2$ and $K/3 < p \leq K$. The polynomial moments satisfy $v_p(\mu(t^e)) \geq -1$. They are integral for $e < 2p - 3$. Indeed, if $p - 1$ divides $2e + 2$, the first three possible multiples of $p - 1$ are canceled respectively by the factors $2e + 3$, $2e + 4$, and $2e + 5$ in (2.2).

Choose $c_e \in \mathbb{Z}_p$ with $c_e \equiv p\mu(t^e) \pmod{p}$, taking $c_e = 0$ for $e < 2p - 3$. Replace each polynomial moment by $\mu_0(t^e) = \mu(t^e) - c_e/p$ and leave the simple-pole values unchanged. The matrix then has the form

$$G_K = A + p^{-1}L, \quad \text{rank}_{\mathbb{Q}_p} L \leq r_p = \max(0, K + 4N - 2p + 2), \quad (4.10)$$

where L is integral. To see the rank bound, the polynomial quotient of Wt^{i+j}/D_{tail} has degree at most $i+j+6N-K$. Hence $L_{ij} = 0$ if $i+j < K-6N+2p-3$. Its first $h-r_p$ rows and columns vanish. Thus the rank bound holds over $\mathbb{Q}_p$, not merely modulo p .

For an ordinary square class a , let $\ell = \ell_K(a)$ and $\delta = \mathbf{1}_{a \leq N}$. There are $\ell - \delta$ remaining poles in the class and exactly $\ell - 2$ of the original poles have $j > p$. Let Q_a be the product of the remaining factors in the class, put $P_a = D_{\text{tail}}/Q_a$, and let E_a be the product of the factors with $j > p$. Use the rows $P_a q_{a,i}$, where

$$q_{a,i}(t) = \begin{cases} (t+a^2)^i, & 0 \leq i < \ell-2, \\ E_a(t)(t+a^2)^{i-(\ell-2)}, & \ell-2 \leq i < \ell-\delta. \end{cases} \quad (4.11)$$

Together with the zero-class rows, these form a unimodular basis. Indeed, the local polynomials are monic of successive degrees, and the resultants of distinct class factors are units.

Cross-class entries for A are integral, since their quotients are polynomials with integral coefficients. Within a class, if either row index is at least $\ell-2$, all poles with $j > p$ cancel. The two possible remaining pole values at a and $p-a$ are integral and congruent modulo p . For their constant coefficients, use $H_{p-a}^{(5)} \equiv H_{a-1}^{(5)} \pmod{p}$ in (2.3). Their coefficients of X are also congruent. The divided difference at the two nodes is therefore integral. Polynomial numerators preserve this congruence.

For two indices $i, j < \ell-2$, the residue part has valuation at least $i+j+6\delta-\ell-4$. The row factors supply $i+j$, the factor W supplies 5δ , the node derivative loses $\ell-\delta-1$, and a harmonic value loses at most 5. The polynomial part is integral. Thus valid nonpositive half-integer weights are

$$w_{a,i} = \begin{cases} \min(0, i+3\delta-(\ell+4)/2), & i < \ell-2, \\ 0, & i \geq \ell-2. \end{cases} \quad (4.12)$$

A single zero-class pole has weight $-1/2$. For two zero-class poles $p, 2p$, the rows $1, t+p^2$ give entry bounds $-3, -1, 0$, so the weights $-2, 0$ are valid.

To restore the correction $p^{-1}L$, we use its rank together with the number of zero row weights. The restriction to nonpositive weights is essential in the following lemma.

Lemma 4.2. *Let A and L be $h \times h$ matrices. Suppose that $v(A_{ij}) \geq w_i + w_j$, where the w_i are nonpositive half-integers. Let r be a nonnegative integer and assume that L is integral with rank at most r . If exactly z weights are zero, then*

$$v \det(A + p^{-1}L) \geq 2 \sum_i w_i - \min(r, z). \quad (4.13)$$

The same assertion holds for the Gauss valuation of polynomial entries.

Proof. Expand the determinant by complementary minors. A term using a k by k minor of L , with row set I and column set J , has valuation at least

$$2 \sum_i w_i - k - \sum_{i \in I} w_i - \sum_{j \in J} w_j.$$

Terms with $k > r$ vanish. If $w_{(1)} \geq \dots \geq w_{(h)}$, the possible loss is at most $\max_{0 \leq k \leq \min(r, h)} (k + 2 \sum_{i=1}^k w_{(i)})$. This expression increases by one at each zero weight and has nonpositive increments after the zero weights. Its maximum is $\min(r, z)$. $\square$

Let $v = K - p \lfloor K/p \rfloor$ and set

$$u = \max(0, N + v - p + 1), \quad t_p = \min(N, v) + u.$$

Summing (4.12) and applying Lemma 4.2 gives the integer

$$\gamma_p^{\text{out}} = \begin{cases} -7(K - p) + 6t_p - 1 - \min(r_p, p - 1 - N + u), & K < 2p, \\ -7(K - p) + 3 + 12N + 5t_p - \min(r_p, p + u), & K \geq 2p. \end{cases} \quad (4.14)$$

For a class containing a removed pole, the counts used in this sum are

ℓ in a removed class	2	3	4	5	6
$-2 \sum_i w_{a,i}$	0	1	2	4	6
$\#\{i \mid w_{a,i} = 0\}$	1	1	2	2	3

An unremoved class has cost $7(\ell - 2)$ and two zero weights. The conditions $a \leq v$ and $a \geq p - v$ among $a \leq N$ give t_p , with overlap u . The zero-class costs are 1 and 4. These counts prove (4.14).

Proposition 4.3. *Under (4.9), $v_p^G(\Delta_K) \geq \gamma_p^{\text{out}}$. For $p > K$, one has $v_p^G(\Delta_K) \geq 0$.*

Proof. The first assertion follows from (4.10) and Lemma 4.2. If $p > K$, every pole is less than p . Each square class contains at most two poles, and the same divided-difference argument makes its contribution integral. The polynomial correction vanishes because $K + 4N + 1 < 2p$. Thus the whole matrix is integral in a unimodular local basis. $\square$

5 Normalization and the prime sum

We combine the local bounds into a positive rational factor $m_{K,M}$ for which $m_{K,M}F_K$ is integral. The exponents must retain their signs. A negative valuation bound requires clearing a denominator, but a positive bound allows division by a power of p . Both contributions enter the growth estimate for $m_{K,M}$.

For $p > K$, put $\gamma_p^{\text{out}} = 0$. For a fixed integer $M \geq 40$ and $K \geq 200M^2$, define

$$L_p(K, M) = \begin{cases} -6h \lfloor \log_p(5K) \rfloor - hv_p(24), & pM \leq K, \\ v_p(S_K) + \gamma_p^{\text{in}}, & pM > K, \ 3p \leq K, \\ v_p(S_K) + \gamma_p^{\text{out}}, & 3p > K. \end{cases} \quad (5.1)$$

Only the primes $p \leq 2h$ enter the normalization

$$m_{K,M} = \prod_{p \leq 2h} p^{-L_p(K,M)}. \quad (5.2)$$

This is a positive rational number. Its exponents are determined by (4.8), (4.14), and Legendre's formula

$$\begin{aligned} v_p(S_K) &= 2h \sum_{a \geq 1} \lfloor K/p^a \rfloor - 12h \sum_{a \geq 1} \lfloor N/p^a \rfloor \\ &\quad - 2 \sum_{i=1}^{h-1} \sum_{a \geq 1} \lfloor 2i/p^a \rfloor + (h-1)v_p(4). \end{aligned} \quad (5.3)$$

Proposition 5.1. *The polynomial $Q_{K,M} = m_{K,M}F_K$ belongs to $\mathbb{Z}[X]$.*

Proof. At every prime $p \leq 2h$, the assertion follows from (3.12) and Propositions 4.1 and 4.3. At $p > 2h$, all factorials in S_K are units, and Δ_K is integral by Proposition 4.3. Thus every coefficient of $Q_{K,M}$ is integral at every prime. $\square$

5.1 The inner limiting function

To estimate $m_{K,M}$, we first express the inner exponents in the variable $x = K/p$. The discrete pole counts become floor functions, and the weighted allocation has the following limiting form.

For $x \geq 3$ and $0 < z < 1/2$, set

$$\ell(x, z) = \lfloor x - z \rfloor + \lfloor x + z \rfloor + 1, \quad b(x, z) = 3\ell(\alpha x, z).$$

In the following formulas,

$$T = \lfloor 2Hx \rfloor, \quad s = Hx - T/2, \quad q = \lfloor 2x \rfloor, \quad n_+ = (2x - q)/2.$$

Define

$$\begin{aligned} \Gamma(x) = & \int_0^{1/2} (T - b(x, z))(T + b(x, z) - \ell(x, z) - 5) dz \\ & + s(2T - q - 5) + (s - n_+)_+. \end{aligned} \quad (5.4)$$

The last two terms allocate the extra rows first to the classes with the larger value of $\ell(x, z)$. Put

$$\begin{aligned} J(u) &= mu - \frac{m(m+1)}{4}, \quad m = \lfloor 2u \rfloor, \\ \mathcal{N}(x) &= 2\lambda x \lfloor x \rfloor - 12\lambda x \lfloor \alpha x \rfloor - 2J(\lambda x), \end{aligned} \quad (5.5)$$

$$R(x) = -\Gamma(x) - \mathcal{N}(x). \quad (5.6)$$

These functions are bounded and piecewise polynomial on each compact subinterval of $[3, \infty)$.

For fixed M , uniformly in the inner range,

$$\gamma_p^{\text{in}} = p\Gamma(K/p) + O_M(1), \quad v_p(S_K) = p\mathcal{N}(K/p) + O_M(1). \quad (5.7)$$

Here and below M remains fixed as K tends to infinity. For the first formula, note that $\ell_A(a) = \ell(A/p, a/p)$. Each interval on which these counts are constant has its number of grid points equal to its length times p , up to an error bounded by two. There are $O(M)$ such intervals, and all dimensions and weights are $O(M)$. The reserved zero block and the change from $p/2$ to $(p-1)/2$ alter the number of allocated rows by $O_M(1)$. Altering that number by one changes the weight sum by $O_M(1)$. If $2Hx$ crosses an integer, allocating one extra row to every ordinary class gives exactly the next base allocation. The estimate is therefore uniform at these transitions as well. The second formula follows from (5.3), since $p^2 > 5K$ and

$$\sum_{i=1}^{h-1} \lfloor 2i/p \rfloor = pJ(h/p) + O_M(1).$$

5.2 The outer limiting function

In the outer range we use $y = p/K$, so that the interval of integration is fixed. We write the residue contribution and the rank correction separately. Define $R_0(y)$ by

$$\begin{cases} 8 - 9y - 8\alpha - 5 \min(\alpha, 1 - 2y) - 5(1 + \alpha - 3y)_+, & 1/3 < y < 1/2, \\ 7(1 - y) - 6 \min(\alpha, 1 - y) - 6(1 + \alpha - 2y)_+ + (1 + 4\alpha - 2y)_+, & 1/2 < y < 1, \\ 0, & y > 1, \end{cases} \quad (5.8)$$

and put

$$d(y) = \left(1 + 4\alpha - 3y - (1 + \alpha - 3y)_+\right)_+ \mathbf{1}_{(1/3, 1/2)}(y). \quad (5.9)$$

Formula (4.14) gives

$$-\gamma_p^{\text{out}} = K(R_0(p/K) - d(p/K)) + O(1).$$

The subtraction of d records the zero-weight restriction in Lemma 4.2. In particular, $\int_{1/3}^{1/2} d(y) dy = 9/640$. The scalar contribution in this range is

$$-v_p(S_K) = K \left(-2\lambda \lfloor 1/y \rfloor + \sum_{j=1}^5 (2\lambda - jy)_+ \right) + O(1).$$

The relevant integral is

$$I_{\text{out}} = \int_{1/3}^{2\lambda} \left(R_0(y) - d(y) - 2\lambda \lfloor 1/y \rfloor + \sum_{j=1}^5 (2\lambda - jy)_+ \right) dy = \frac{127751}{96000}. \quad (5.10)$$

An exact evaluation is given in Appendix B.

The prime number theorem now converts the local bounds into integrals. We use only its asymptotic form [9, §27.12].

Proposition 5.2. *For each fixed integer $M \geq 40$,*

$$\limsup_{\substack{K \rightarrow \infty \\ 40|K}} \frac{\log m_{K,M}}{K^2} \leq I_{\text{out}} + \frac{6\lambda}{M} + \int_3^M \frac{R(x)}{x^3} dx. \quad (5.11)$$

Proof. For $p \leq \sqrt{5K}$, (3.12) contributes $O(K^{3/2} \log K) = o(K^2)$. For $\sqrt{5K} < p \leq K/M$, its logarithmic factor is one. The prime number theorem gives the contribution $6\lambda K^2/M + o(K^2)$.

For a bounded piecewise continuous function f on $[3, M]$, partial summation and the prime number theorem give

$$\frac{1}{K^2} \sum_{K/M < p \leq K/3} pf(K/p) \log p \longrightarrow \int_3^M \frac{f(x)}{x^3} dx.$$

Use (5.7) with $f = R$. The $O_M(1)$ errors contribute $O_M(K) = o(K^2)$. The same argument in the variable $y = p/K$ gives (5.10) for the outer range. This proves (5.11). $\square$

5.3 The tail integral

We split the inner integral at 20. Its bounded part is evaluated in Appendix B. For the tail, write $R(x)$ as a linear term with periodic coefficient plus a bounded remainder. This gives explicit bounds at both 20 and the cutoff M .

Let $f = \{x\}$, $g = \{\alpha x\}$, $\tau = \{2Hx\}$, $\sigma = \{2x\}$, and $\eta = \{2\lambda x\}$. Set

$$A(z) = \ell(x, z) - 2x, \quad B(z) = \ell(\alpha x, z) - 2\alpha x.$$

Expanding (5.4) and (5.5) gives

$$R(x) = xF(x) + Q(x), \quad F(x) = 4\lambda + 2\lambda f - 12\lambda g, \quad (5.12)$$

$$Q(x) = \frac{\tau(\tau - \sigma) - (\tau - \sigma)_+ + \eta(1 - \eta)}{2} + 9 \int_0^{1/2} B(z)^2 dz - 3 \int_0^{1/2} A(z)B(z) dz. \quad (5.13)$$

Both A and B have mean zero. Each squared integral is at most $1/8$, and Cauchy–Schwarz bounds the absolute value of the mixed integral by $1/8$. The first numerator in (5.13) lies between $-1/4$ and $1/4$. Consequently

$$-\frac{1}{2} \leq Q(x) \leq \frac{13}{8}. \quad (5.14)$$

Define the continuous periodic functions

$$\begin{aligned} \mathcal{P}(x) &= 74g(1 - g) - \lambda f(1 - f), & \bar{\mathcal{P}} &= \frac{2923}{240}, \\ G_0(v) &= \frac{v(1 - v)(2v - 1)}{6}, & \mathcal{C}(x) &= \frac{74}{\alpha} G_0(g) - \lambda G_0(f). \end{aligned}$$

Away from their finitely many breakpoints in a period,

$$\mathcal{P}' = F + \lambda, \quad \mathcal{C}' = \mathcal{P} - \bar{\mathcal{P}}.$$

The identity $\max_{[0,1]} |G_0| = 1/(36\sqrt{3})$ implies

$$|\mathcal{C}(x)| \leq \frac{118511}{4320\sqrt{3}} < 16.$$

Two integrations by parts now give

$$\begin{aligned} \int_T^\infty \frac{R(x)}{x^3} dx &= -\frac{\lambda}{T} - \frac{\mathcal{P}(T)}{T^2} + \frac{\bar{\mathcal{P}}}{T^2} - \frac{2\mathcal{C}(T)}{T^3} \\ &\quad + 6 \int_T^\infty \frac{\mathcal{C}(x)}{x^4} dx + \int_T^\infty \frac{Q(x)}{x^3} dx. \end{aligned} \quad (5.15)$$

At $T = 20$, one has $\mathcal{P}(T) = 37/2$ and $\mathcal{C}(T) = 0$. Using (5.14) gives

$$\int_{20}^\infty \frac{R(x)}{x^3} dx \leq -\frac{2689}{48000}. \quad (5.16)$$

If $40 \mid M$, then $\mathcal{P}(M) = \mathcal{C}(M) = 0$, so

$$\int_M^\infty \frac{R(x)}{x^3} dx \geq -\frac{\lambda}{M} + \frac{2923/240 - 1/4}{M^2} - \frac{32}{M^3}. \quad (5.17)$$

Appendix B proves

$$\int_3^{20} \frac{R(x)}{x^3} dx = \frac{322437603634266857629}{7535670527041937280000}. \quad (5.18)$$

Put

$$A_* = \frac{9928298118277006344769}{7535670527041937280000}, \quad (5.19)$$

$$A_M = A_* + \frac{7\lambda}{M} - \frac{2923/240 - 1/4}{M^2} + \frac{32}{M^3}. \quad (5.20)$$

Combining (5.11) with (5.16)–(5.18) proves

$$\limsup_{\substack{K \rightarrow \infty \\ 40|K}} K^{-2} \log m_{K,M} \leq A_M \quad (M \in 40\mathbb{Z}_{>0}). \quad (5.21)$$

6 The real determinant

We next bound $F_K(\zeta(5))$. The positive moment representation expresses $\Delta_K(\zeta(5))$ as a multiple integral with a squared Vandermonde factor. On scaling $y_i = K\sqrt{t_i}$, the exponent is a discrete logarithmic energy in the field V below. We bound it for every configuration by comparison with an explicit measure of mass $h/K = \lambda$.

Define the external field on $[0, \infty)$ by

$$V(t) = 2\pi\sqrt{t} + \int_0^1 \log(t + u^2) du - 6 \int_0^\alpha \log(t + u^2) du. \quad (6.1)$$

For a measure ρ , write

$$U^\rho(t) = \int \log|t - u| d\rho(u), \quad I(\rho) = \iint \log|t - u| d\rho(t) d\rho(u).$$

We use the same notation for measures on $\mathbb{C}$ whenever the integrals are absolutely convergent.

Lemma 6.1. *There is a positive measure ρ of mass λ , supported in $(0, 2)$, such that*

$$2U^\rho(t) - V(t) \leq M_0 = -\frac{1329}{200} \quad (t \geq 0). \quad (6.2)$$

It is a sum of sixteen arcsine measures on nested intervals, each of length greater than $1/225$. Moreover, if

$$C_* = -2\lambda + 12\alpha\lambda(1 - \log \alpha) + 3\lambda^2 - 2\lambda^2 \log(2\lambda), \quad (6.3)$$

then

$$\lambda M_0 - I(\rho) + C_* \leq \bar{U} = -\frac{2733991}{2000000} < -\frac{136699}{100000}. \quad (6.4)$$

Appendix A gives the measure and proves these bounds. The argument needs a comparison measure, not an explicit solution of the equilibrium problem. The lower bound for the interval lengths controls the error in the regularization below.

6.1 A bound for every configuration

The comparison rests on negativity of logarithmic energy for a measure of mass zero. We include a proof with the integrability hypothesis needed for the circle and arcsine measures in our application.

Lemma 6.2 (Zero-mass logarithmic energy). *Let ν be a finite real signed Borel measure on $\mathbb{C}$ with compact support. Suppose that $\nu(\mathbb{C}) = 0$ and*

$$\iint_{\mathbb{C} \times \mathbb{C}} |\log |z - w|| \, d|\nu|(z) \, d|\nu|(w) < \infty,$$

where $|\nu|$ is the total variation measure. Then

$$I(\nu) \leq 0. \tag{6.5}$$

Proof. For $s > 0$, completing the square gives

$$e^{-s|z-w|^2} = \frac{4s}{\pi} \int_{\mathbb{C}} e^{-2s|z-u|^2} e^{-2s|w-u|^2} \, dA(u),$$

where A is planar Lebesgue measure. Fubini's theorem therefore yields

$$J(s) := \iint e^{-s|z-w|^2} \, d\nu(z) \, d\nu(w) = \frac{4s}{\pi} \int_{\mathbb{C}} \left(\int e^{-2s|z-u|^2} \, d\nu(z) \right)^2 \, dA(u) \geq 0.$$

These interchanges are justified by the bounded Gaussian kernel and the finiteness of $|\nu|$.

For $0 < a < b < \infty$ and $r > 0$, put

$$L_{a,b}(r) = \frac{1}{2} \int_a^b \frac{e^{-s} - e^{-sr^2}}{s} \, ds.$$

The identity

$$\frac{1}{2} \int_0^\infty \frac{e^{-s} - e^{-sr^2}}{s} \, ds = \log r$$

follows by writing $e^{-s} - e^{-sr^2} = \int_1^{r^2} s e^{-sv} \, dv$ and integrating first in s . The integrand has a fixed sign for each r , so $|L_{a,b}(r)| \leq |\log r|$. Also, $L_{a,b}(r) \rightarrow \log r$ as $a \downarrow 0$ and $b \uparrow \infty$. The mass-zero condition gives

$$\iint L_{a,b}(|z - w|) \, d\nu(z) \, d\nu(w) = -\frac{1}{2} \int_a^b \frac{J(s)}{s} \, ds \leq 0.$$

The logarithmic integrability assumption implies that the diagonal has zero $|\nu| \otimes |\nu|$ measure. Dominated convergence now proves (6.5). $\square$

We apply the lemma after regularizing a configuration $t_1, \dots, t_h$ in $[0, \infty)$. Let ω_i be normalized arclength on the circle of radius $\varepsilon > 0$ centered at t_i in $\mathbb{C}$, and set $\sigma = K^{-1} \sum_{i=1}^h \omega_i$. Since $h/K = \lambda$, the signed measure $\sigma - \rho$ has mass zero.

The circle and arcsine measures have continuous finite logarithmic potentials. Hence the potential of the positive measure $\eta = \sigma + \rho$ is bounded on its compact support. The positive part of the logarithmic kernel is bounded there as well, so $\iint |\log |z - w|| \, d\eta(z) \, d\eta(w) < \infty$. Since $|\sigma - \rho| \leq \eta$, Lemma 6.2 applies.

A unit arcsine measure on an interval of length L assigns at most $4\sqrt{2r}/(\pi\sqrt{L})$ to the intersection with any interval of radius r . The circle-average identity [11, Example 1.10] gives $\int \log |z - u| d\omega_i(z) = \log \max(|t_i - u|, \varepsilon)$. Integrating the preceding mass bound in r therefore gives

$$\int U^\rho d\omega_i - U^\rho(t_i) \leq \sum_j \frac{8c_j\sqrt{2\varepsilon}}{\pi\sqrt{b_j - a_j}} \leq 60\sqrt{\varepsilon}.$$

The same circle-average identity gives $I(\omega_i) = \log \varepsilon$. For the mutual energy, applying it successively to the two circles gives

$$\iint \log |z - w| d\omega_i(z) d\omega_j(w) \geq \int \log |t_i - w| d\omega_j(w) = \log \max(|t_i - t_j|, \varepsilon) \geq \log |t_i - t_j|.$$

Expanding $I(\sigma - \rho) \leq 0$ now yields

$$\begin{aligned} 2 \sum_{i < j} \log |t_i - t_j| - K \sum_i V(t_i) &\leq (\lambda M_0 - I(\rho))K^2 \\ &\quad + 120\lambda K^2\sqrt{\varepsilon} - h \log \varepsilon. \end{aligned} \quad (6.6)$$

The inequality includes coincident points by interpreting their left-hand side as $-\infty$.

A uniform bound on the exponent is not enough on an unbounded domain. We retain the integrable factor $e^{-\sqrt{t}}$ by changing the field from $V(t)$ to $V(t) - \sqrt{t}/K$. For $K \geq 2$,

$$2U^\rho(t) - V(t) + \frac{\sqrt{t}}{K} \leq M_0 + \frac{\sqrt{2}}{K}. \quad (6.7)$$

For $t \leq 2$, this follows from (6.2). For $t \geq 2$, use the support of ρ and $\log(1 + u) \leq u$ to obtain

$$2U^\rho(t) - V(t) \leq \frac{13}{10} \log t + \frac{6\alpha^3}{t} - 2\pi\sqrt{t}. \quad (6.8)$$

After adding $\sqrt{t}/K$, the right-hand side is decreasing on $[2, \infty)$ and its value at 2 is less than M_0 . These statements already follow from $\log 2 < 7/10$, $\pi > 3$, and $\sqrt{2} > 7/5$. Apply (6.6) with (6.7) and $\varepsilon = K^{-2}$. The result is

$$\begin{aligned} 2 \sum_{i < j} \log |t_i - t_j| - K \sum_i V(t_i) + \sum_i \sqrt{t_i} &\leq (\lambda M_0 - I(\rho))K^2 \\ &\quad + (120 + \sqrt{2})h + 2h \log K. \end{aligned} \quad (6.9)$$

6.2 The Gram integral and scaling

Andréief's integration identity [7, (1.7)], applied to the moment matrix in (2.4), gives

$$\Delta_K(\zeta(5)) = \frac{1}{h!} \int_{(0, \infty)^h} \prod_{i < j} (y_i^2 - y_j^2)^2 \prod_{i=1}^h \frac{D_N(y_i^2)^6}{D_K(y_i^2)} w(y_i) dy_i. \quad (6.10)$$

The identity also follows by expanding the two Vandermonde determinants and integrating term by term. Absolute convergence follows from the exponential decay of w . The elementary bound

$$w(y) \leq 8192(1 + y)^5 e^{-2\pi y} \quad (y > 0) \quad (6.11)$$

follows from $\sum_{\ell \geq 1} \ell^4 q^\ell = q(1 + 11q + 11q^2 + q^3)/(1 - q)^5$ and $1 - e^{-u} \geq u/(1 + u)$.

For $1 \leq m \leq K$ and $t \geq 0$, monotonicity gives

$$0 \leq \sum_{j=1}^m \log(t + (j/K)^2) - K \int_0^{m/K} \log(t + u^2) du \leq 2 \log K + 2. \quad (6.12)$$

The difference decreases with t , since $u \mapsto (t + u^2)^{-1}$ is decreasing. At $t = 0$ the bound is the elementary factorial estimate $\log(m!) \leq m \log m - m + \log m + 1$.

In (6.10), put $y_i = K\sqrt{t_i}$. Combining (6.11) and (6.12) gives

$$\begin{aligned} \Delta_K(\zeta(5)) &\leq \frac{K^{2h(h+6N-K)+16h}(4096e^{12})^h}{h!} \\ &\quad \times \int_{(0,\infty)^h} \prod_{i < j} (t_i - t_j)^2 \prod_i e^{-KV(t_i)} t_i^{-1/2} (1 + \sqrt{t_i})^5 dt_i. \end{aligned} \quad (6.13)$$

Since

$$\int_0^\infty t^{-1/2} (1 + \sqrt{t})^5 e^{-\sqrt{t}} dt = 652,$$

the remaining integral is bounded by applying (6.9) and then integrating $\prod_i t_i^{-1/2} (1 + \sqrt{t_i})^5 e^{-\sqrt{t_i}}$. Taking logarithms and enlarging the linear term gives

$$\begin{aligned} \log \Delta_K(\zeta(5)) &\leq 2h(h + 6N - K) \log K + (\lambda M_0 - I(\rho))K^2 \\ &\quad + 18h \log K + 160h. \end{aligned} \quad (6.14)$$

The factorial inequalities

$$m \log m - m + 1 \leq \log(m!) \leq m \log m - m + \log m + 2$$

and summation of $\log((2i)!)$ give

$$-2 \sum_{i=1}^{h-1} \log((2i)!) \leq -2h^2 \log(2h) + 3h^2 + 4h \log(2h).$$

It follows from (2.5) that

$$\begin{aligned} \log S_K &\leq (2\lambda - 12\alpha\lambda - 2\lambda^2)K^2 \log K + C_* K^2 \\ &\quad + 6h \log K + 6h. \end{aligned} \quad (6.15)$$

The $K^2 \log K$ terms in (6.14) and (6.15) cancel. The remaining quadratic coefficient is $\lambda M_0 - I(\rho) + C_*$, bounded by $\bar{U}$ in Lemma 6.1. We have proved the following estimate.

Proposition 6.3. *For every $K \in 40\mathbb{Z}_{>0}$,*

$$0 < F_K(\zeta(5)), \quad \log F_K(\zeta(5)) \leq \bar{U} K^2 + 24K \log K + 200K. \quad (6.16)$$

7 Completion of the irrationality proof

It remains to compare the normalizing factor with the real determinant. Both estimates have been obtained on the K^2 scale, with M fixed.

Proof of Theorem 2.1. Proposition 5.1 gives $Q_{K,M} \in \mathbb{Z}[X]$. Its degree is h by (2.9), and its value at $\zeta(5)$ is positive by Proposition 2.2 and the positivity of the normalizing factors. Moreover, (5.21) and Proposition 6.3 give

$$\limsup_{\substack{K \rightarrow \infty \\ 40|K}} K^{-2} \log Q_{K,M}(\zeta(5)) \leq A_M + \bar{U} \quad (M \in 40\mathbb{Z}_{>0}). \quad (7.1)$$

The rational constants satisfy

$$-1600(A_{200} + \bar{U}) > \frac{139}{5}, \quad -1600(A_{100000} + \bar{U}) > \frac{7907}{100}. \quad (7.2)$$

Their exact positive margins are given in Appendix B. Substituting $K = 40n$ proves (2.7) and (2.8). $\square$

Proof of Theorem 1.1. Apply Theorem 2.1 with $M = 200$. If $\zeta(5) = a/b$ with $a \in \mathbb{Z}$ and $b \in \mathbb{Z}_{>0}$, then $b^{37n} Q_{40n,200}(a/b)$ is a positive integer that tends to zero, by (2.7). This is impossible. $\square$

A The measure and its elementary bounds

This appendix proves Lemma 6.1. We specify the sixteen arcsine components, bound their potential on a rational partition, and evaluate the energy using the nesting of the intervals. The same elementary enclosures give the constant needed in Proposition C.1.

A.1 The sixteen intervals

Let $\omega_{[a,b]}$ be the probability measure on (a, b) with density

$$\frac{1}{\pi \sqrt{(t-a)(b-t)}}.$$

Define

$$\rho = \sum_{j=1}^{16} c_j \omega_{[a_j, b_j]},$$

where Table 1 gives $10^{12}a_j$, $10^{12}b_j$, and $10^{12}c_j$. All entries in that table are integers. Direct summation gives $\sum c_j = 37/40$. The intervals satisfy

$$0 < a_{16} < \cdots < a_1 < b_1 < \cdots < b_{16} < 2, \quad b_j - a_j > 1/225.$$

The potential of a unit arcsine measure is given by the classical interval formula [11, Example 1.11], with our sign convention,

$$U^{\omega_{[a,b]}}(t) = \begin{cases} \log((b-a)/4), & a \leq t \leq b, \\ \log \frac{|t - (a+b)/2| + \sqrt{(t-a)(t-b)}}{2}, & t \notin [a, b]. \end{cases} \quad (\text{A.1})$$

Table 1: The rational arcsine measure.

j	$10^{12}a_j$	$10^{12}b_j$	$10^{12}c_j$
1	3906748086	8992695531	10515596180
2	2312248264	15340997855	29471737793
3	1402286665	25730180724	42934365099
4	881725356	41909578246	58204231966
5	578197906	65851089563	69037621310
6	396324613	99481037884	78873099189
7	283911191	144325727458	84856120711
8	212206188	201105762729	88396082127
9	165097686	269345996903	88303382125
10	133347132	347089554156	85472321255
11	111522114	430806704415	78899184238
12	96349355	515561896511	70353471918
13	85815639	595448778546	58838976615
14	78667711	664241383483	44421321106
15	74129565	716160577112	30462865791
16	71741310	746637295669	5959622577

For completeness, write $m = (a + b)/2$ and $r = (b - a)/2$ and substitute $u = m + r \cos \theta$ in the defining integral. For $t > b$, differentiation gives $((t - a)(t - b))^{-1/2}$. Integration, with the normalization $U(t) - \log t \rightarrow 0$ as $t \rightarrow \infty$, gives the second branch. Reflection treats $t < a$. On the interval, the substitution $t = m + r \cos \varphi$ and the identity $\cos \varphi - \cos \theta = -2 \sin((\varphi + \theta)/2) \sin((\varphi - \theta)/2)$ give the constant $\log(r/2)$. Continuity includes the endpoints.

Put $S_j = \sum_{i \leq j} c_i$ and $S_0 = 0$. Nesting makes the potential of the j th interval constant on every earlier interval. Therefore

$$I(\rho) = \sum_{j=1}^{16} (S_j^2 - S_{j-1}^2) \log \frac{b_j - a_j}{4}. \quad (\text{A.2})$$

A.2 Elementary enclosures

We use rational enclosures for logarithms, arctangents, and square roots. The formulas and truncation orders are fixed here so that each subsequent inequality is determined by the displayed rational data. For $1 \leq x \leq 2$, put $z = (x - 1)/(x + 1)$. Then

$$0 \leq \log x - 2 \sum_{k=0}^{m-1} \frac{z^{2k+1}}{2k+1} \leq \frac{2z^{2m+1}}{(2m+1)(1-z^2)}. \quad (\text{A.3})$$

For a general positive argument, use $x = 2^e r$ with $1 \leq r < 2$ and $\log x = e \log 2 + \log r$. For $0 \leq z \leq 1/2$,

$$\left| \arctan z - \sum_{k=0}^{m-1} \frac{(-1)^k z^{2k+1}}{2k+1} \right| \leq \frac{z^{2m+1}}{2m+1}. \quad (\text{A.4})$$

Larger arguments are reduced by

$$\arctan x = \frac{\pi}{2} - \arctan(1/x) \quad (x > 1), \quad \arctan x = 2 \arctan \frac{x}{1 + \sqrt{1+x^2}} \quad (0 \leq x \leq 1),$$

and $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$. A rational square root is enclosed by two adjacent multiples of 2^{-144} , whose squares bracket its argument. In (A.3) take $m = 64$, and in (A.4) take $m = 80$. The remainder bounds on the reduced ranges are less than 2^{-144} . These prescriptions give rational upper and lower bounds for every expression below.

A.3 The potential inequality on the half-line

For $t > 0$, integration in (6.1) gives

$$V(t) = \log(1+t) - 6\alpha \log(t + \alpha^2) - 2 + 12\alpha + 2\sqrt{t} \left(\pi + \arctan \frac{1}{\sqrt{t}} - 6 \arctan \frac{\alpha}{\sqrt{t}} \right). \quad (\text{A.5})$$

At zero, $V(0) = -12\alpha \log \alpha - 2 + 12\alpha$. Writing $\Phi(y) = V(y^2)$ gives

$$\Phi'(y) = 2 \left(\pi + \arctan(1/y) - 6 \arctan(\alpha/y) \right), \quad \Phi''(y) = -\frac{2}{1+y^2} + \frac{12\alpha}{\alpha^2 + y^2}.$$

The second derivative changes sign once, at $y^2 = 711/880$. Also $\Phi'(0+) = -3\pi$ and $\Phi'(+\infty) = 2\pi$. It follows that V has exactly one minimum. Put

$$q_- = \frac{59205077}{10^{10}}, \quad q_+ = \frac{59205079}{10^{10}}.$$

The inequalities (A.4) give $\Phi'(\sqrt{q_-}) < 0 < \Phi'(\sqrt{q_+})$. Thus the minimum lies between q_- and q_+ .

Because the intervals are nested, U^ρ is nonincreasing on $[0, a_1]$, constant on $[a_1, b_1]$, and nondecreasing on $[b_1, \infty)$. Its maximum on any interval is attained at an endpoint. For $q_- \leq t \leq q_+$, a lower bound for (A.5) is

$$V_* = \log(1 + q_-) - 6\alpha \log(q_+ + \alpha^2) - 2 + 12\alpha + 2\sqrt{q_+} \left(\pi + \arctan \frac{1}{\sqrt{q_+}} - 6 \arctan \frac{\alpha}{\sqrt{q_+}} \right). \quad (\text{A.6})$$

The parenthesis is negative. Each factor in (A.5) is bounded in the direction shown, which proves this lower bound. For an interval $[l, r] \subset [0, 2]$, set

$$\mathcal{B}(l, r) = 2 \max(U^\rho(l), U^\rho(r)) - \begin{cases} V(r), & r \leq q_-, \\ V(l), & l \geq q_+, \\ V_*, & \text{otherwise.} \end{cases} \quad (\text{A.7})$$

The preceding monotonicity bounds give $2U^\rho(t) - V(t) \leq \mathcal{B}(l, r)$ for $l \leq t \leq r$.

The following partition supplies the intervals in (A.7). Define

$$A_0 = 0, \quad A_j = a_{17-j} \quad (1 \leq j \leq 16), \quad A_{17} = q_-, \quad A_{18} = q_+, \\ A_{18+j} = b_j \quad (1 \leq j \leq 16), \quad A_{35} = 2.$$

A row (j, d, k) in Table 2 denotes

$$\left[A_j + (A_{j+1} - A_j) \frac{k}{2^d}, A_j + (A_{j+1} - A_j) \frac{k+1}{2^d} \right]. \quad (\text{A.8})$$

A range of integers in the last column includes both endpoints. These intervals partition $[0, 2]$. Substitution of (A.1) and (A.5) into (A.7), with (A.3) and (A.4), gives the single rational bound

$$\mathcal{B}(l, r) < -\frac{6645002}{10^6} < -\frac{1329}{200} \quad (\text{A.9})$$

for every interval in the table. All arguments and endpoints needed in this inequality are specified by Table 1 and (A.8).

Table 2: Intervals for the potential bound.

j	d	k
0	1	0
0	2	2, 3
1	0	0
2	0	0
3	0	0
4	0	0
5	0	0
6	0	0
7	0	0
8	0	0
9	1	0, 1
10	1	0, 1
11	1	0, 1
12	1	0
12	2	2, 3
13	2	0, ..., 3
14	2	0, ..., 3
15	1	0
15	2	2, 3
16	0	0
17	0	0
18	0	0
19	1	1
19	3	0 2, 3
19	4	2, 3
20	2	3
20	3	4, 5
20	4	0, 1 6, 7
20	5	4, ..., 11
21	2	3
21	3	5
21	4	0 7, ..., 9
21	5	2, ..., 4 10, ..., 13
21	6	10, ..., 19
22	3	5, ..., 7
22	4	0 8, 9
22	5	2, 3 12, ..., 15
22	6	8, ..., 23
23	3	6, 7
23	4	9, ..., 11
23	5	0, ..., 2 14, ..., 17
23	6	6, ..., 13 19, ..., 27

Continued on the next page

Table 2 continued

j	d	k
23	7	28, ..., 37
24	3	6, 7
24	4	10, 11
24	5	0, ..., 2 15, ..., 19
24	6	6, ..., 11 22, ..., 29
24	7	24, ..., 43
25	3	7
25	4	10, ..., 13
25	5	0, ..., 2 16, ..., 19
25	6	6, ..., 11 24, ..., 31
25	7	24, ..., 47
26	3	7
26	4	11, ..., 13
26	5	0, ..., 2 17, ..., 21
26	6	6, ..., 11 25, ..., 33
26	7	24, ..., 49
27	4	11, ..., 15
27	5	0, ..., 2 17, ..., 21
27	6	6, ..., 11 26, ..., 33
27	7	24, ..., 51
28	4	12, ..., 15
28	5	0, ..., 2 18, ..., 23
28	6	6, ..., 11 27, ..., 35
28	7	24, ..., 53
29	4	13, ..., 15
29	5	0, ..., 2 19, ..., 25
29	6	6, ..., 12 27, ..., 37
29	7	26, ..., 53
30	4	13, ..., 15
30	5	0, ..., 2 20, ..., 25
30	6	6, ..., 14 26, ..., 39
30	7	30, ..., 51
31	4	14, 15
31	5	0, ..., 2 20, ..., 27
31	6	6, ..., 39
32	4	15
32	5	0, ..., 4 19, ..., 29
32	6	10, ..., 37
33	4	0 15
33	5	2, ..., 29
34	2	2, 3
34	3	3
34	4	3, ..., 5
34	5	4, 5
34	6	5, ..., 7
34	7	6, ..., 9
34	8	7, ..., 11
34	9	5, ..., 13
34	10	0, ..., 9

For $t \geq 2$, (6.8) is decreasing and its value at 2 is less than $-1329/200$. This proves (6.2) on the whole half-line.

A.4 Energy and the norm constant

Applying (A.3) to (A.2) and (6.3) gives

$$\begin{aligned} -\frac{2126593445148}{10^{12}} &< I(\rho) < -\frac{2126593445147}{10^{12}}, \\ \frac{2653035990340}{10^{12}} &< C_* < \frac{2653035990341}{10^{12}}. \end{aligned} \quad (\text{A.10})$$

Since $\lambda M_0 = -49173/8000$, these bounds imply

$$\lambda M_0 - I(\rho) + C_* < -\frac{1366995564511}{10^{12}} < -\frac{2733991}{2000000}.$$

This completes the proof of Lemma 6.1. The same elementary bounds in (C.6) give

$$\frac{2773171335773}{200000000000} < \kappa < \frac{6932928339433}{500000000000} < \frac{6933}{500}. \quad (\text{A.11})$$

B Exact arithmetic integrals

We evaluate the two bounded integrals in the prime sum and give the exact margins in (7.2). The inner numerator $R(x)$ and the outer integrand are affine on their respective rational intervals.

B.1 Evaluation of the inner integral

The functions in (5.6) admit a formula using only fractional parts and minima. For $0 \leq u < 1$, set

$$d_0(u) = \min(u, 1 - u), \quad e(u) = \begin{cases} 1, & u \leq 1/2, \\ -1, & u > 1/2. \end{cases}$$

The floor-function formula for ℓ makes sense for every real x . For such x , put $f = \{x\}$. Then, for almost every $z \in (0, 1/2)$,

$$\ell(x, z) - 2x = e(f)(\mathbf{1}_{z < d_0(f)} - 2d_0(f)).$$

The only possible exception is $z = d_0(f)$, which does not affect integration. Applying the same identity to αx , with $g = \{\alpha x\}$, gives

$$\begin{aligned} \int_0^{1/2} B(z)^2 dz &= d_0(g)(1 - 2d_0(g)), \\ \int_0^{1/2} A(z)B(z) dz &= e(f)e(g)(\min(d_0(f), d_0(g)) - 2d_0(f)d_0(g)). \end{aligned} \quad (\text{B.1})$$

Equations (5.12), (5.13), and (B.1) are an explicit rational formula for $R(x)$.

Let $3 = t_0 < t_1 < \dots < t_s = 20$ be the ordered set consisting of the endpoints and all $x \in (3, 20)$ for which cx is an integer for some

$$c \in \{2, 2\alpha, 2\lambda, 2H, 4\alpha, 2(1 - \alpha), 2(1 + \alpha)\}. \quad (\text{B.2})$$

There are $s = 143$ intervals. On each open interval (t_i, t_{i+1}) , all fractional-part choices and all branches of the minima in (B.1) are fixed. The possible extra changes occur when $d_0(f) = d_0(g)$ or $\tau = \sigma$. They are included in (B.2). The quadratic coefficient in R cancels, since

$$2\lambda - 12\lambda\alpha + 2(H^2 - H - \lambda^2) - 18\alpha^2 + 6\alpha = 0.$$

Thus R is affine on every such interval.

For $l = t_i$, $r = t_{i+1}$, put

$$x_i = (2l + r)/3, \quad y_i = (l + 2r)/3, \quad a_i = \frac{R(y_i) - R(x_i)}{y_i - x_i}, \quad b_i = R(x_i) - a_i x_i.$$

The exact integral is therefore the rational sum

$$\int_3^{20} \frac{R(x)}{x^3} dx = \sum_{i=0}^{142} \left[a_i \left(\frac{1}{t_i} - \frac{1}{t_{i+1}} \right) + \frac{b_i}{2} \left(\frac{1}{t_i^2} - \frac{1}{t_{i+1}^2} \right) \right]. \quad (\text{B.3})$$

Grouping its terms into unit intervals gives Table 3. Their sum is (5.18).

Table 3: The inner integral on unit intervals.

j	$\int_j^{j+1} R(x)x^{-3} dx$
3	$\frac{26807}{161280}$
4	$\frac{37383}{704000}$
5	$\frac{37523}{8236800}$
6	$-\frac{120923}{6552000}$
7	$-\frac{10025233}{356428800}$
8	$-\frac{110029309}{3348864000}$
9	$-\frac{2278419487}{64465632000}$
10	$-\frac{282415081}{7724640000}$
11	$-\frac{3236921227}{87524236800}$
12	$-\frac{3350220001}{90899827200}$
13	$\frac{7424224373}{2217983040000}$
14	$\frac{16775764609}{955086612480}$
15	$\frac{369043847}{30810528000}$
16	$\frac{651380108633}{86461373856000}$
17	$\frac{472851276229}{119820121344000}$
18	$\frac{4930060867}{4724197793280}$
19	$-\frac{15199801}{11563552000}$

B.2 Evaluation of the outer integral

Recall the rank correction $d(y)$ from (5.9), which is distinct from d_0 . The outer integrand is

$$T(y) = R_0(y) - d(y) - 2\lambda \lfloor 1/y \rfloor + \sum_{j=1}^5 (2\lambda - jy)_+.$$

On the intervals in Table 4, write $T(y) = b + cy$. The displayed coefficients follow by resolving the positive parts in (5.8). Integrating $b + cy$ over these intervals gives $I_{\text{out}} = 127751/96000$.

Table 4: The affine pieces of the outer integrand.

l	r	b	c
$\frac{1}{3}$	$\frac{43}{120}$	$\frac{279}{40}$	-9
$\frac{43}{120}$	$\frac{37}{100}$	$\frac{451}{40}$	-21
$\frac{37}{100}$	$\frac{13}{30}$	$\frac{377}{40}$	-16
$\frac{13}{30}$	$\frac{37}{80}$	$\frac{429}{40}$	-19
$\frac{37}{80}$	$\frac{1}{2}$	$\frac{17}{4}$	-5
$\frac{1}{2}$	$\frac{43}{80}$	$\frac{51}{10}$	-3
$\frac{43}{80}$	$\frac{37}{60}$	$\frac{231}{20}$	-15
$\frac{37}{60}$	$\frac{13}{20}$	$\frac{97}{10}$	-12
$\frac{13}{20}$	$\frac{37}{40}$	$\frac{42}{5}$	-10
$\frac{37}{40}$	1	1	-2
1	$\frac{37}{20}$	$\frac{37}{20}$	-1

B.3 The final rational margins

The values of (5.20) at the two cutoffs are

$$A_{200} = \frac{127125602969131786927559}{94195881588024216000000}, \quad A_{100000} = \frac{7756864096839411316755964319057}{5887242599251513500000000000000}.$$

With $\bar{U} = -2733991/2000000$, the exact positive differences in (7.2) are

$$\begin{aligned} -1600(A_{200} + \bar{U}) - \frac{139}{5} &= \frac{3089837638249482469}{58872425992515135000}, \\ -1600(A_{100000} + \bar{U}) - \frac{7907}{100} &= \frac{29873543950273155160680943}{3679526624532195937500000000}. \end{aligned}$$

In particular $A_{100000} + \bar{U} < -79/1600$, as required in (C.7).

C Rational approximation and coefficient height

We prove Corollary 1.2 by controlling the determinant near $\zeta(5)$. The additional input is a relative matrix norm estimate, combined with the stronger decay in (2.8). We then record a coefficient height bound. None of these additional estimates is needed for Theorem 1.1.

C.1 The relative matrix norm

A polynomial that is small and nonzero at $\zeta(5)$ may vanish at a nearby rational number. To exclude this possibility, we use the affine dependence of $G_K(X)$ on X and measure its change relative to the positive matrix $G_K(\zeta(5))$.

Put $\xi = \zeta(5)$, $G = G_K(\xi)$, and $M_K = [X]G_K(X)$. Since G is positive definite, the real symmetric matrix

$$B_K = G^{-1/2} M_K G^{-1/2}$$

is defined. Its norm is the Euclidean operator norm.

Proposition C.1. *For all sufficiently large $K \in 40\mathbb{Z}_{>0}$,*

$$\|B_K\| \leq e^{\varrho K}, \quad \varrho = \frac{6933}{500}. \quad (\text{C.1})$$

Proof. Let q be a real polynomial of degree less than h , and write $p(t) = q(K^2 t)$ and $\mathcal{A}_K = K^{12N-2K+6}$. Fix $0 < a < b$. Keeping only the term $\ell = 1$ in (2.10), restricting the integral to $y = K\sqrt{t}$ with $a \leq t \leq b$, and bounding every factor separately gives

$$q^\top G q \geq \mathcal{A}_K c_a e^{-\mathcal{F}(a,b)K} \int_a^b p(t)^2 dt, \quad (\text{C.2})$$

where q on the left denotes its coefficient vector and

$$c_a = \frac{(2\pi)^4 a^2}{24}, \quad \mathcal{F}(a, b) = 2\pi\sqrt{b} + \log(1+b) - 6\alpha \log a.$$

Indeed, the scaled numerator is at least a^{6N} , the denominator is at most $(1+b)^K$, and the remaining density is at least $c_a e^{-2\pi K\sqrt{b}}$.

Set $u_j = j/K$. The residue expression for M_K gives

$$q^\top M_K q = \mathcal{A}_K \sum_{j=N+1}^K r_j p(-u_j^2)^2, \quad r_j = \frac{u_j^4 \prod_{\ell=1}^N (u_\ell^2 - u_j^2)^5}{\prod_{\substack{N < k \leq K \\ k \neq j}} (u_k^2 - u_j^2)}. \quad (\text{C.3})$$

The numerator has absolute value at most one. Since $u_k + u_j \geq 2\alpha$,

$$\begin{aligned} |r_j| &\leq \frac{(K/(2\alpha))^{h-1}}{(j-N-1)!(K-j)!} \leq \frac{(K/\alpha)^{h-1}}{(h-1)!} \\ &\leq \exp\{\lambda K(1 - \log(\alpha\lambda))\}. \end{aligned} \quad (\text{C.4})$$

For the last inequality, use $m! \geq (m/e)^m$ and the fact that $x \mapsto x \log(eK/(\alpha x))$ increases for $0 < x \leq h$.

The Legendre expansion of p on $[a, b]$ gives

$$\max_{-1 \leq t \leq 0} |p(t)|^2 \leq \frac{h^2 \mathcal{R}^{2h}}{b-a} \int_a^b p(t)^2 dt, \quad Y = \frac{2+a+b}{b-a}, \quad \mathcal{R} = Y + \sqrt{Y^2 - 1}. \quad (\text{C.5})$$

For completeness, the transformed argument has absolute value at most Y . The Laplace integral for P_j gives $|P_j(s)| \leq \mathcal{R}^j$ there [9, (18.10.5)]. Cauchy–Schwarz in the orthogonal expansion, followed by $\sum_{j=0}^{h-1} (2j+1) = h^2$, proves (C.5).

Combining (C.2)–(C.5) yields

$$\|B_K\| \leq \frac{h^3}{c_a(b-a)} e^{\kappa(a,b)K}, \quad \kappa(a, b) = \mathcal{F}(a, b) + \lambda(1 - \log(\alpha\lambda)) + 2\lambda \log \mathcal{R}.$$

Take $a = 1/18$ and $b = 1/3$. Then $Y = 43/5$ and

$$\begin{aligned} \kappa &= \frac{2\pi}{\sqrt{3}} + \log \frac{4}{3} + \frac{9}{20} \log 18 + \frac{37}{40} \left(1 + \log \frac{1600}{111} \right) \\ &\quad + \frac{37}{20} \log \frac{43 + 4\sqrt{114}}{5} < \frac{6933}{500}. \end{aligned} \tag{C.6}$$

The strict inequality is established in Appendix A. Its positive margin absorbs the polynomial factor in h , proving (C.1). $\square$

C.2 Proof of Corollary 1.2

Choose K proportional to $\log b$. The relative norm bound will preserve positivity, while the quadratic decay will outweigh the factor b^h . Use $M = 100000$ and write $Q_K = Q_{K,100000}$. Equation (7.1) implies, for all sufficiently large allowed K ,

$$0 < Q_K(\xi) \leq e^{-\epsilon K^2}, \quad \epsilon = \frac{79}{1600}. \tag{C.7}$$

Let $c = 75/4$. For a positive integer b , choose

$$K = 40 \left\lceil \frac{c \log b}{40} \right\rceil, \quad c \log b \leq K < c \log b + 40.$$

Suppose that $a \in \mathbb{Z}$ and $|a/b - \xi| \leq b^{-260}$. Set $\delta = a/b - \xi$. The choice of constants gives

$$\varrho c - 260 = -\frac{1}{80}, \quad \epsilon - \frac{\lambda}{c} = \frac{1}{24000}.$$

Hence, by Proposition C.1,

$$|\delta| \|B_K\| \leq e^{40\varrho} b^{-1/80} = o(K^{-1}).$$

Thus all eigenvalues of $I + \delta B_K$ are positive for sufficiently large b . Also $\det(I + \delta B_K) \leq (1 + |\delta| \|B_K\|)^h \leq \exp(h|\delta| \|B_K\|) \rightarrow 1$, so the determinant is at most two. The affine dependence of G_K on X gives

$$Q_K(a/b) = Q_K(\xi) \det(I + \delta B_K).$$

Since $Q_K \in \mathbb{Z}[X]$ has degree h , it follows that

$$0 < b^h Q_K(a/b) \leq 2 \exp(\lambda K \log b - \epsilon K^2) \leq 2e^{-K^2/24000} < 1$$

for all sufficiently large b . This contradicts integrality and proves Corollary 1.2. The threshold is uniform in a .

C.3 Coefficient height and primitive normalization

The relative matrix estimate also controls the coefficients, without any further local argument. We retain $M = 100000$ in the height bound and then compare the explicit normalization with the primitive one.

Let $H(Q)$ be the largest absolute value of a coefficient of Q . If $\beta_1, \dots, \beta_h$ are the eigenvalues of B_K , then

$$Q_K(X) = Q_K(\xi) \prod_{j=1}^h (1 + (X - \xi)\beta_j).$$

The elementary bound $\xi < 2$ and Proposition C.1 imply

$$\log H(Q_K) \leq (\lambda_\varrho - \epsilon)K^2 + o(K^2) = \frac{511067}{40000}K^2 + o(K^2) < 13K^2 \quad (\text{C.8})$$

for all sufficiently large allowed K .

The cutoff enters the explicit normalizing factor, but it can be removed by passing to the primitive polynomial. For $0 \neq A \in \mathbb{Q}[X]$, let $\text{cont}_{\mathbb{Q}}(A) > 0$ be the rational number such that $A/\text{cont}_{\mathbb{Q}}(A)$ has integral coefficients with greatest common divisor one. Put

$$P_K = \Delta_K / \text{cont}_{\mathbb{Q}}(\Delta_K).$$

For every admissible K, M , there is a positive integer $a_{K,M}$ with $Q_{K,M} = a_{K,M}P_K$. Indeed, a rational multiple of a primitive integer polynomial has integer coefficients only when the multiplier is an integer. Thus $P_K(\xi) > 0$, and the upper bounds proved above for $Q_{K,M}(\xi)$ hold for $P_K(\xi)$ as well. This is a consequence of the explicit normalization, rather than a second irrationality argument.

Author's note on further work

The author is working on extensions to higher odd zeta values and has made several advances in this direction. Some of the local functional estimates and determinant arguments appear to extend beyond the present case. The construction for $\zeta(5)$ also relies on special features of its positive moment representation and arithmetic cancellations. Obtaining compatible positivity and denominator estimates for higher odd values remains a central part of this work.

These investigations also concern linear independence among specified odd zeta values. The author is exploring related questions about the transcendence of $\zeta(5)$ and selected families of odd zeta values, as well as algebraic independence involving $\zeta(5)$. Developments outside the argument presented here suggest possible approaches to some of these questions, although their scope remains unsettled. No such conclusion follows from the irrationality theorem or the approximation bound in this paper. A general transcendence result for odd zeta values would require further ideas.

The author is also actively developing alternative approaches to the irrationality of $\zeta(5)$ and has made substantial progress in this work. These include variants of the present construction and approaches based on different proof methods. These investigations suggest that the present construction may not be the only route to the result. The proof given here arose from simplifying and strengthening earlier approaches developed by the author, and further simplifications or sharper estimates may be possible. A related aim is to isolate general principles linking local determinant divisibility, positivity, and real decay.

These remarks describe the author's ongoing investigations. The advances mentioned above are not developed in this paper, and no extension or alternative proof is claimed here.

Use of generative AI

A generative AI tool was used in a supporting role for editing the exposition, proofreading, L^AT_EX preparation, and consistency checks of the arguments, calculations, and references. The author takes full responsibility for the content and correctness of this work.

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